

IIT HYDERABAD

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices and Circuits Laboratory Report

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1 Objective

- To assemble a simple two-transistor MOSFET differential amplifier using one SI9926BDY dual N-channel MOSFET package.
- To set and verify the DC operating point of the differential amplifier.
- To experimentally obtain the common-mode transfer characteristic and input common-mode range (ICMR).
- To experimentally obtain the differential-mode transfer characteristic (ΔI_D versus V_{id}).
- To compare the measured small-signal transconductance with the theoretical prediction.

2 Components and Equipment Required

1. SI9926BDY dual N-channel MOSFET

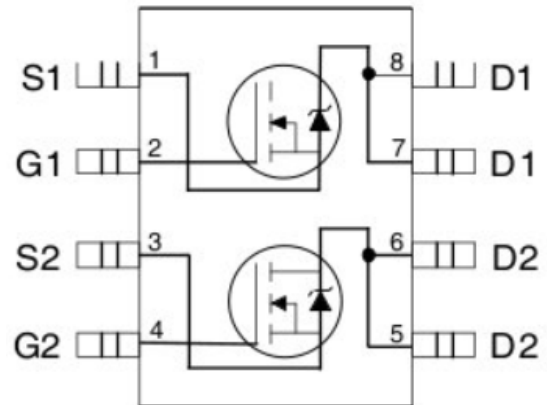
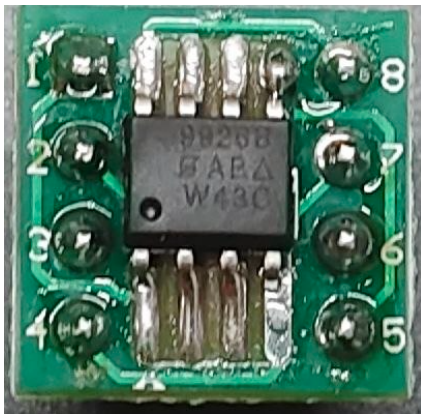


Figure 1: SI9926BDY SO-8

2. Resistors ($R_{D1} = R_{D2} = 180 \, \Omega$, $R_{SS} = 22.4 \, \Omega$)
3. DC power supply
4. Digital multimeter
5. Breadboard and jumper wires

3 Device Characterstics of Matched Dual Mosfets

Prior to assembling the differential amplifier, the input and output static characteristics of both transistors in the dual SI9926 package (designated Top Transistor M_1 and Bottom Transistor M_2) were experimentally recorded.

Table 1: Output Characteristics ($V_{GS} = 0.6$ V)

MOSFET M_1		MOSFET M_2	
V_{DS} (V)	I_D (mA)	V_{DS} (V)	I_D (mA)
0.00	0.0000	0.00	0.0000
0.20	0.0128	0.20	0.0123
0.40	0.0140	0.40	0.0132
0.60	0.0149	0.60	0.0143
0.80	0.0158	0.80	0.0151
1.00	0.0165	1.00	0.0160
1.20	0.0172	1.20	0.0165
1.40	0.0180	1.40	0.0171
1.60	0.0187	1.60	0.0178
1.80	0.0194	1.80	0.0184
2.00	0.0200	2.00	0.0192
4.00	0.0265	4.00	0.0250
6.00	0.0330	6.00	0.0311
8.00	0.0405	8.00	0.0379

Table 2: Input/Transfer Characteristics ($V_{DS} = 5$ V)

MOSFET M_1		MOSFET M_2	
V_{GS} (V)	I_D (mA)	V_{GS} (V)	I_D (mA)
0.50	0.0031	0.50	0.0033
0.60	0.0301	0.60	0.0307
0.61	0.0390	0.61	0.0384
0.62	0.0487	0.62	0.0486
0.63	0.0620	0.63	0.0613
0.64	0.0790	0.64	0.0764
0.65	0.100	0.65	0.098
0.66	0.128	0.66	0.126
0.67	16.4	0.67	16.63
0.68	16.70	0.68	16.89
0.69	16.74	0.69	16.90
0.70	16.77	0.70	16.93
0.80	17.60	0.80	17.05
1.00	220.0	1.00	240.0

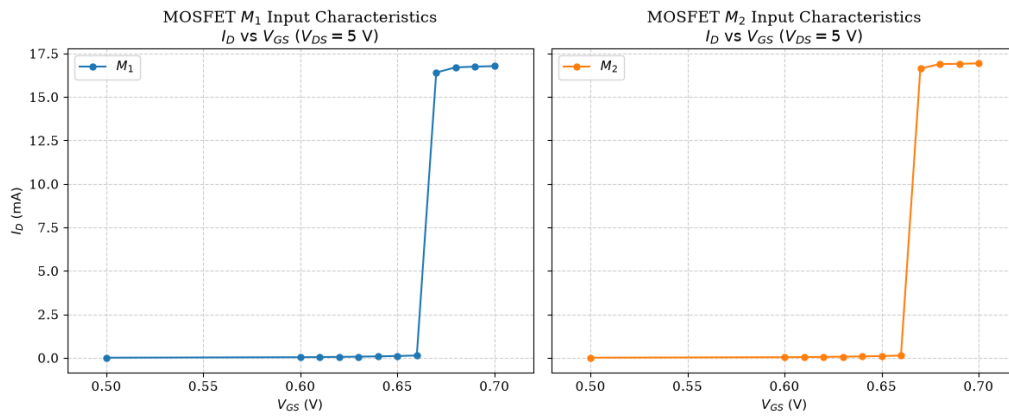


Figure 2: Input Characterstics of two MOSFETs

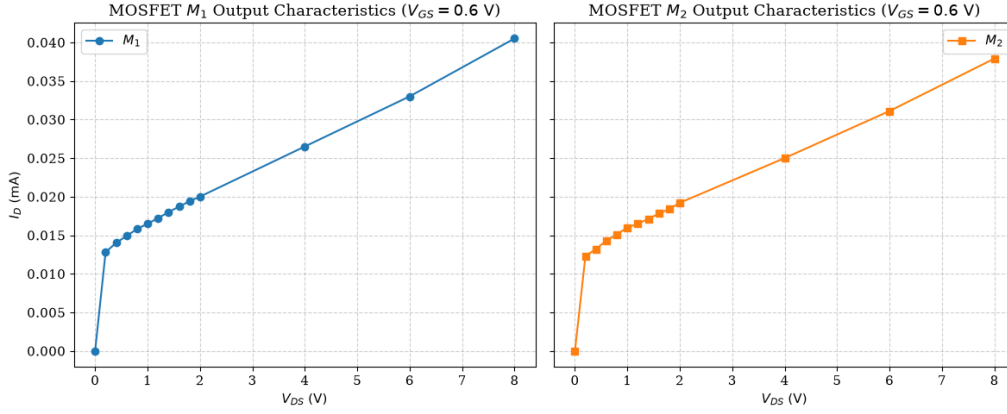


Figure 3: Output Characteristics of two MOSFETs

4 DC Quiescent (Q) Bias Setup and Verification

4.1 Theoretical Background

The differential pair consists of two matched n-channel MOSFETs (M_1 and M_2) sharing a common source node, with $V_{in1} = V_{in2} = V_Q = 2.71$ V. The quiescent gate bias is selected to establish a nominal tail current of $I_{SS} \approx 80$ mA, with $I_{D1,Q} \approx I_{D2,Q} \approx 40$ mA per branch, defining the operating Q-point.

The drain currents are given by

$$I_{D1} = \frac{V_{DD} - V_{D1}}{R_D}, \quad I_{D2} = \frac{V_{DD} - V_{D2}}{R_D}, \quad I_{SS} = I_{D1} + I_{D2}. \quad (1)$$

At balanced operation ($V_{id} = 0$),

$$I_{D1,Q} = I_{D2,Q} = \frac{I_{SS}}{2}, \quad V_{D1,Q} = V_{D2,Q} = V_{DD} - \frac{I_{SS}R_D}{2}. \quad (2)$$

For saturation operation,

$$V_{DS} \geq V_{GS} - V_{TH}. \quad (3)$$

The quiescent bias establishes the operating point for differential amplification with sufficient voltage headroom for signal variation.

4.2 Q-Point Observation

Table 3: Measured DC Quiescent (Q-Point) Parameters

Parameter / Node	Symbol	Measured Value
Supply Voltage	V_{DD}	12.00 V
Drain Load Resistance	R_D	180.0 Ω
Gate 1 DC Bias Voltage	$V_{in1,Q}$	2.710 V
Gate 2 DC Bias Voltage	$V_{in2,Q}$	2.710 V
Differential Input Voltage	$V_{id} = V_{in1} - V_{in2}$	0.000 V
Drain 1 Voltage	$V_{D1,Q}$	4.580 V
Drain 2 Voltage	$V_{D2,Q}$	4.440 V
Quiescent Drain Difference	$\Delta V_{D,Q} = V_{D1} - V_{D2}$	0.140 V

4.3 Calculations and Verification

1. Tail Current

From the tail resistor $R_{SS} = 22.4 \Omega$ and $V_S = 1.80$ V,

$$I_{SS} = \frac{V_S}{R_{SS}} = \frac{1.80}{22.4} \approx 80.36 \text{ mA}. \quad (4)$$

2. Quiescent Drain Currents

Using $R_D = 180 \Omega$,

$$I_{D1,Q} = \frac{12.00 - 4.580}{180} \approx 41.22 \text{ mA}, \quad (5)$$

$$I_{D2,Q} = \frac{12.00 - 4.440}{180} \approx 42.00 \text{ mA}. \quad (6)$$

Thus,

$$I_{D1,Q} + I_{D2,Q} = 83.22 \text{ mA}, \quad (7)$$

which differs from the measured tail current by approximately 3.56%.

3. Symmetry and Saturation Verification

The ideal branch current is

$$I_{D,\text{ideal}} = \frac{I_{SS}}{2} = 40.18 \text{ mA}, \quad (8)$$

giving a branch current mismatch of approximately 1.94% and a drain voltage offset of

$$|V_{D1,Q} - V_{D2,Q}| = 140 \text{ mV}. \quad (9)$$

The quiescent voltages are

$$V_{GS} = 2.71 - 1.80 = 0.91 \text{ V}, \quad (10)$$

$$V_{DS1} = 4.580 - 1.80 = 2.780 \text{ V}, \quad (11)$$

$$V_{DS2} = 4.440 - 1.80 = 2.640 \text{ V}. \quad (12)$$

Since both devices have adequate V_{DS} and positive drain-to-gate voltage, they operate in saturation at the Q-point.

5 Common-Mode Operation

5.1 Theoretical Background and ICMR Definition

The Input Common-Mode Range (ICMR) is the span of input voltages ($V_{in+} = V_{in-} = V_{cm}$) over which both MOSFETs (M_1, M_2) remain biased in active saturation while the tail element (R_{SS}) functions properly. Outside this range, linear amplification fails.

1. Lower Boundary ($V_{cm,min}$)

Conduction requires overcoming the gate threshold and supporting the tail voltage:

$$V_{cm,min} \approx V_{th} + V_{RSS} \quad (13)$$

For $V_{cm} < V_{cm,min}$, the devices enter **Cutoff** ($V_{GS} < V_{th}$):

- Currents collapse: $I_{D1} = I_{D2} \approx 0$ and $I_S \approx 0$.
- Drains pull up to the supply rail: $V_{D1} \approx V_{D2} \approx V_{DD} = 12 \text{ V}$.

2. Upper Boundary ($V_{cm,max}$)

Maintaining saturation requires $V_{DS} \geq V_{GS} - V_{th}$, which simplifies to:

$$V_{cm,max} \leq V_D + V_{th} \quad (14)$$

As V_{cm} rises, increasing drain current drives $V_D = V_{DD} - I_D R_D$ downward. Exceeding $V_D + V_{th}$ forces the transistors into the **Triode (Ohmic)** region, drastically degrading transconductance and gain.

5.2 Circuit Diagram

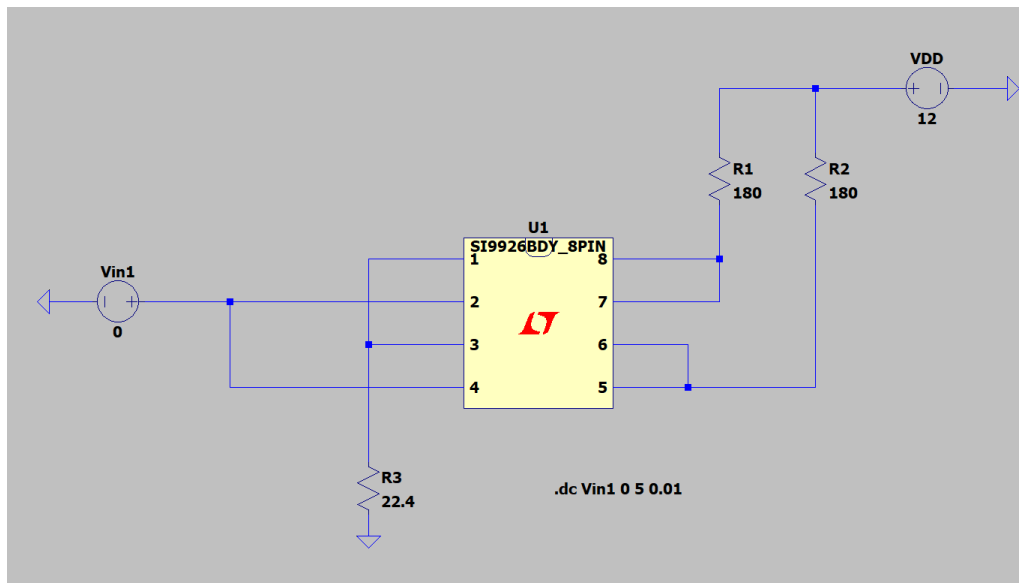


Figure 4: LTspice Schematic

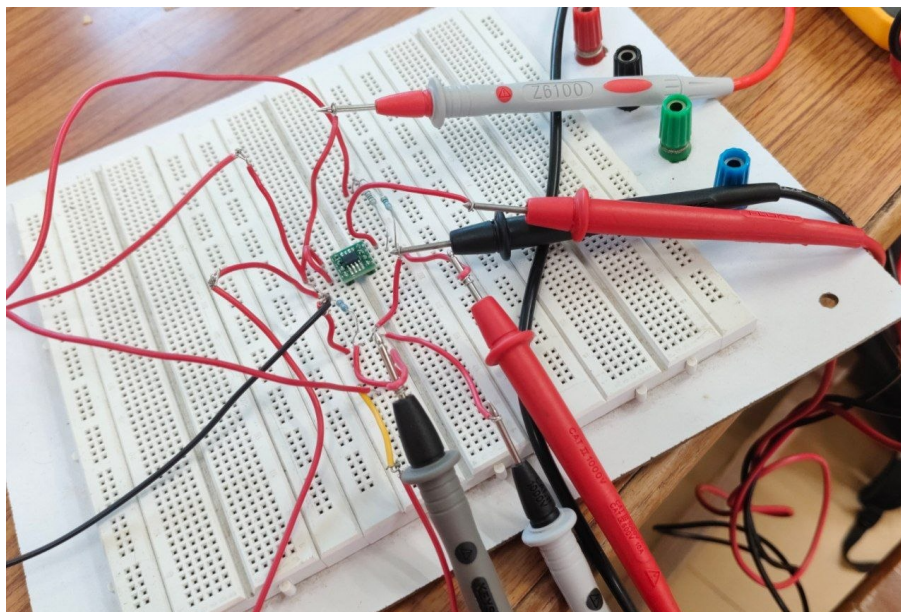


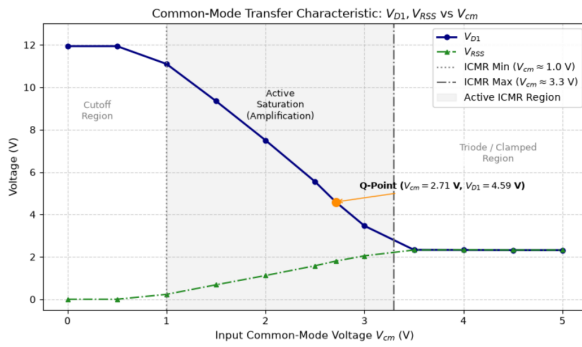
Figure 5: Circuit

5.3 Observation Table : Common-Mode

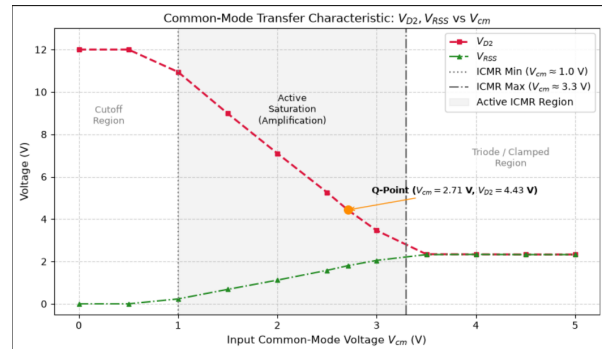
Table 4: Common-mode transfer characteristic measurements ($V_{DD} = 12\text{ V}$, $R_{D1} = R_{D2} = 180\ \Omega$, $R_{SS} = 22.4\ \Omega$)

V_{cm} (V)	V_{D1} (V)	V_{D2} (V)	V_{RSS} (V)	$I_S = \frac{V_{RSS}}{R_{SS}}$ (mA)
0.00	11.94	12.00	0.000001	0.00
0.50	11.94	12.00	0.00045	0.02
1.00	11.10	10.94	0.232	10.36
1.50	9.35	8.98	0.683	30.49
2.00	7.50	7.10	1.120	50.00
2.50	5.55	5.26	1.580	70.54
2.71	4.59	4.43	1.800	80.36
3.00	3.463	3.460	2.050	91.52
3.50	2.332	2.343	2.323	103.71
4.00	2.326	2.336	2.332	104.11
4.50	2.318	2.330	2.3230	103.71
5.00	2.317	2.329	2.3233	103.72

5.4 Graphical Analysis : Transfer Curves and ICMR Demarcation

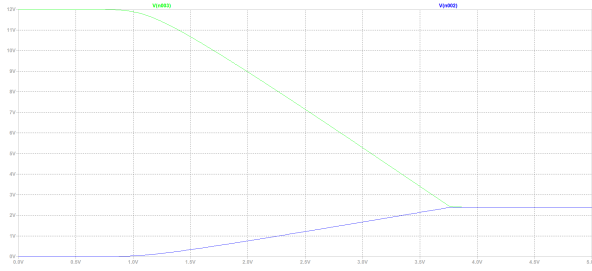
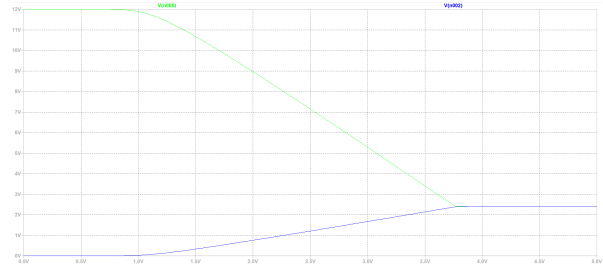
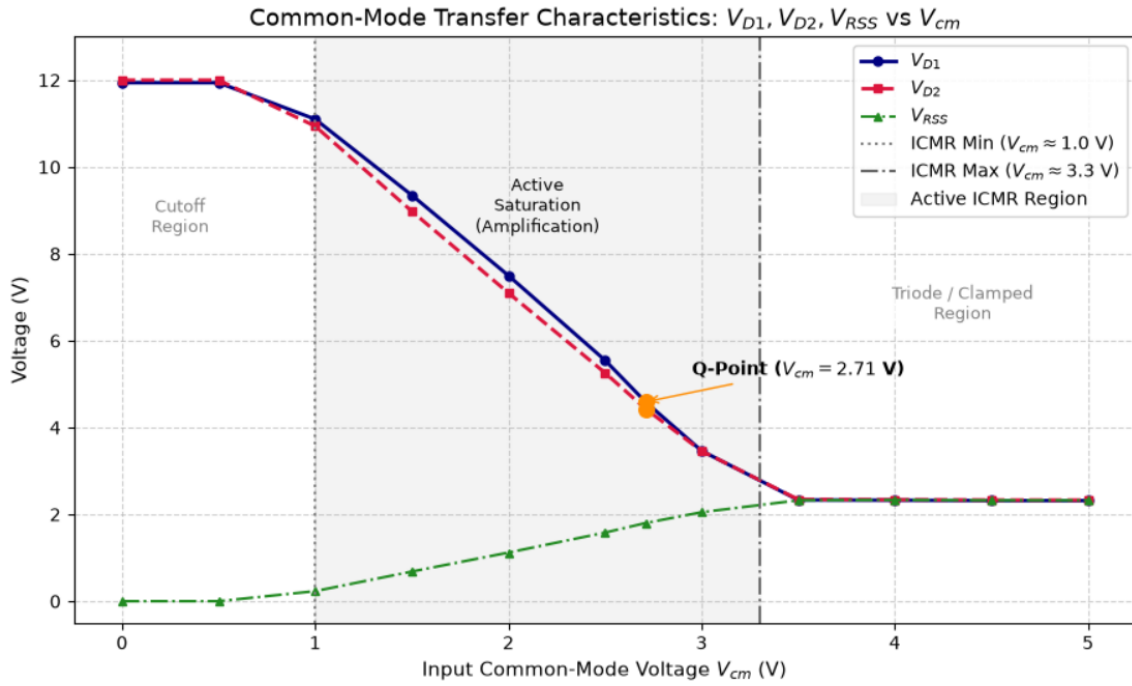
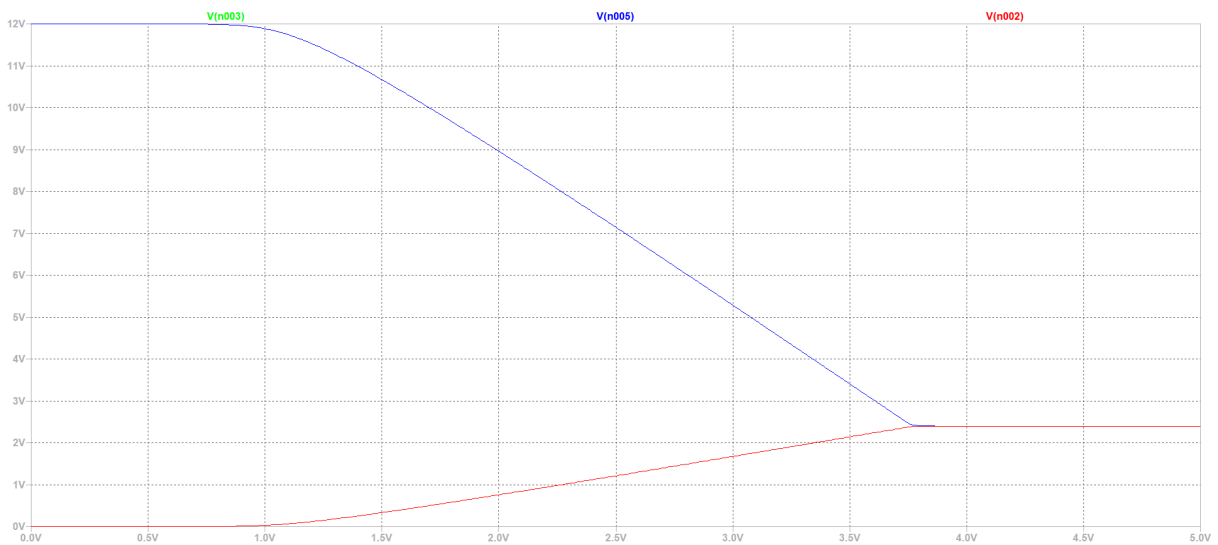


(a) V_{D1} and V_{RSS} versus V_{cm}



(b) V_{D2} and V_{RSS} versus V_{cm}

Figure 6: Experimental Plots

(a) Simulated V_{D1} and V_{RSS} versus V_{cm} (b) Simulated V_{D2} and V_{RSS} versus V_{cm} **Figure 7:** Simulation Plots**Figure 8:** Experimental Plot**Figure 9:** Simulation Plot

5.5 Calculations and Results

1. Input Common-Mode Range (ICMR)

From the observation table and transfer characteristics :

- **Lower Limit ($V_{cm,min}$):** Conduction begins as V_{cm} overcomes the threshold at $V_{cm,min} \approx 1.00$ V. Below this, the pair is in cutoff with $V_D \approx 12.0$ V and $I_S \approx 0$ mA.
- **Upper Limit ($V_{cm,max}$):** Saturation ends near $V_{cm,max} \approx 3.30$ V as the transistors enter triode. At $V_{cm} \geq 3.50$ V, drain voltages collapse into the tail node and clamp at ≈ 2.33 V.
- **Demarcated Range:** ICMR = [1.00 V, 3.30 V].

6 Differential-Mode Operation

6.1 Theoretical Background: Weak-Inversion Exponential Model

In differential-mode operation, equal and opposite input signals are applied to the two MOSFET gates such that

$$V_{G1} = V_Q + \frac{V_{id}}{2}, \quad V_{G2} = V_Q - \frac{V_{id}}{2}.$$

The differential input voltage steers the tail current between the two MOSFETs while the total current remains approximately constant. The differential drain current is theoretically given by

$$\Delta I_D = I_{D1} - I_{D2} = I_{SS} \tanh \left(\frac{V_{id}}{2nV_T} \right).$$

For small differential inputs, the characteristic is approximately linear, and its slope represents the small-signal transconductance g_m .

6.2 Circuit Diagram

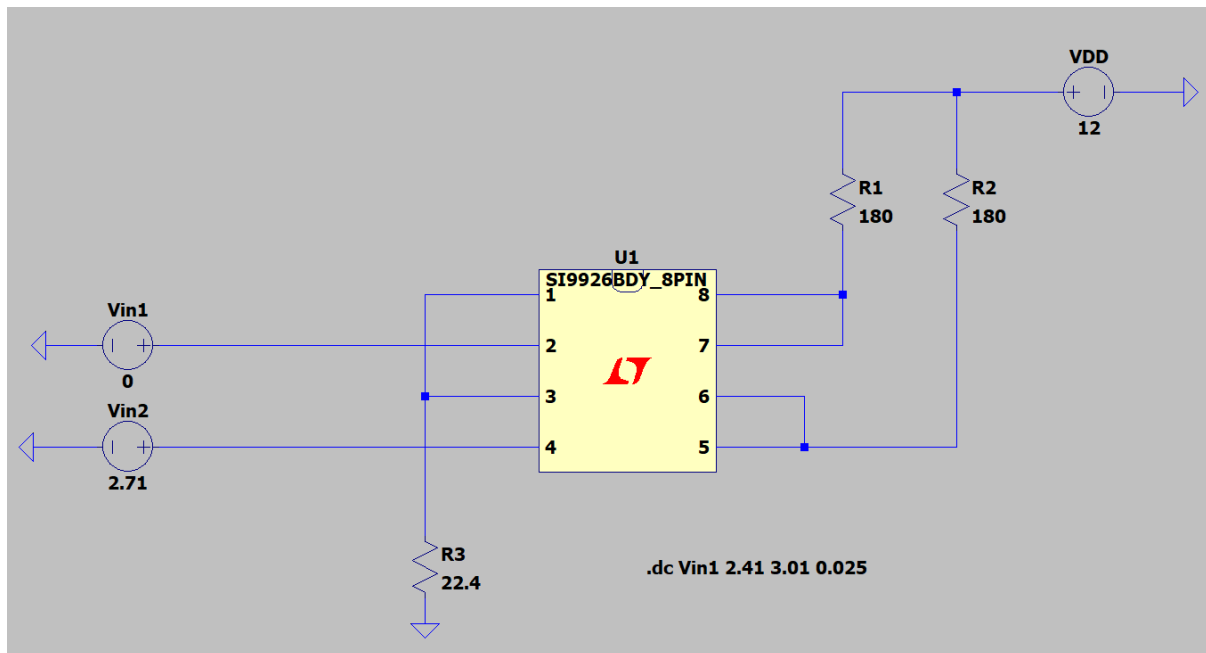


Figure 10: LTspice Schematic

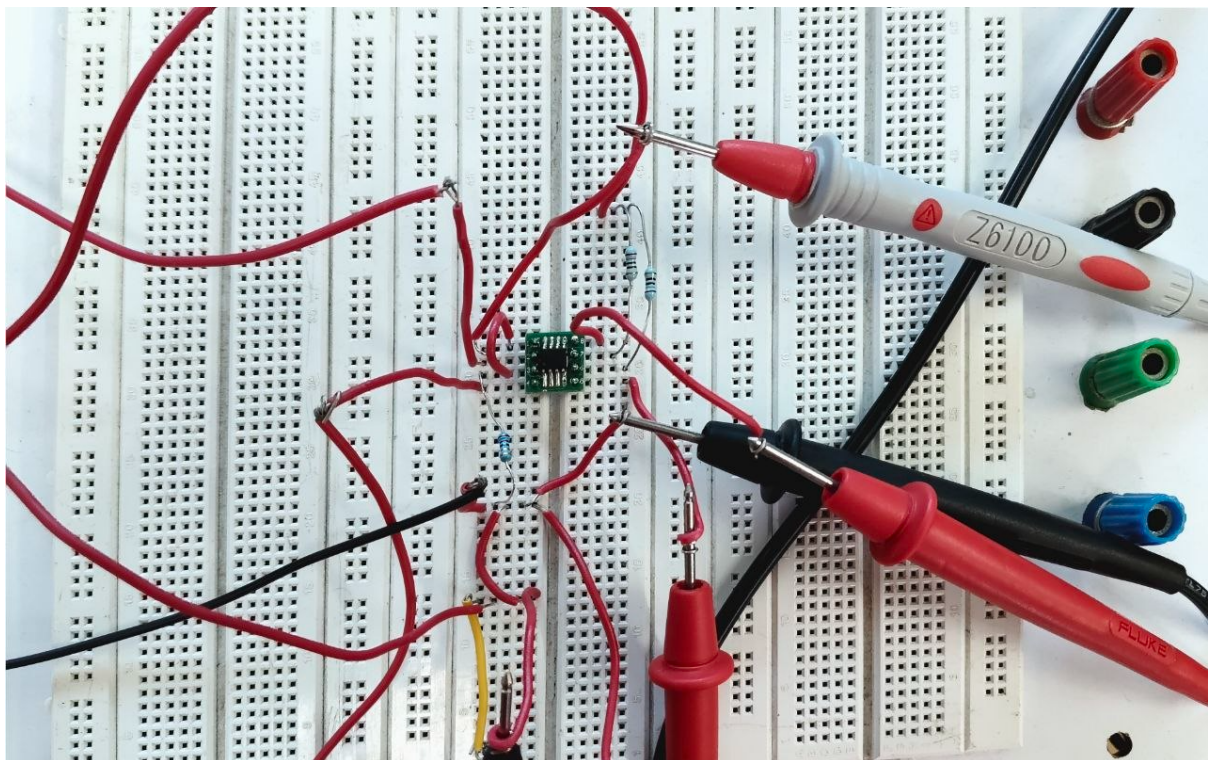


Figure 11: Circuit

6.3 Observation Table: Differential Sweep

Table 5: Measured Differential Sweep Characteristics ($V_{DD} = 12\text{ V}$, $R_D = 180\ \Omega$, $V_Q = 2.71\text{ V}$)

$V_{in1}\text{ (V)}$	$V_{id}\text{ (mV)}$	$V_{D1}\text{ (V)}$	$V_{D2}\text{ (V)}$	$I_{D1}\text{ (mA)}$	$I_{D2}\text{ (mA)}$	$\Delta I_D\text{ (mA)}$	$ \Delta I_D \text{ (mA)}$
2.41	-300	9.210	1.660	15.50	57.44	-41.94	41.94
2.46	-250	8.830	1.700	17.61	57.22	-39.61	39.61
2.51	-200	8.460	1.728	19.67	57.07	-37.40	37.40
2.56	-150	8.020	1.819	22.11	56.56	-34.45	34.45
2.61	-100	7.220	2.310	26.56	53.83	-27.28	27.28
2.66	-50	5.960	3.310	33.56	48.28	-14.72	14.72
2.71	0	4.580	4.440	41.22	42.00	-0.78	0.78
2.76	+50	3.260	5.590	48.56	35.61	+12.94	12.94
2.81	+100	2.187	6.540	54.52	30.33	+24.18	24.18
2.86	+150	1.908	6.780	56.07	29.00	+27.07	27.07
2.91	+200	1.874	6.830	56.26	28.72	+27.53	27.53
2.96	+250	1.861	6.840	56.33	28.67	+27.66	27.66
3.01	+300	1.853	6.850	56.37	28.61	+27.76	27.76

6.4 Graphical Analysis

6.4.1 Differential Drain Voltages

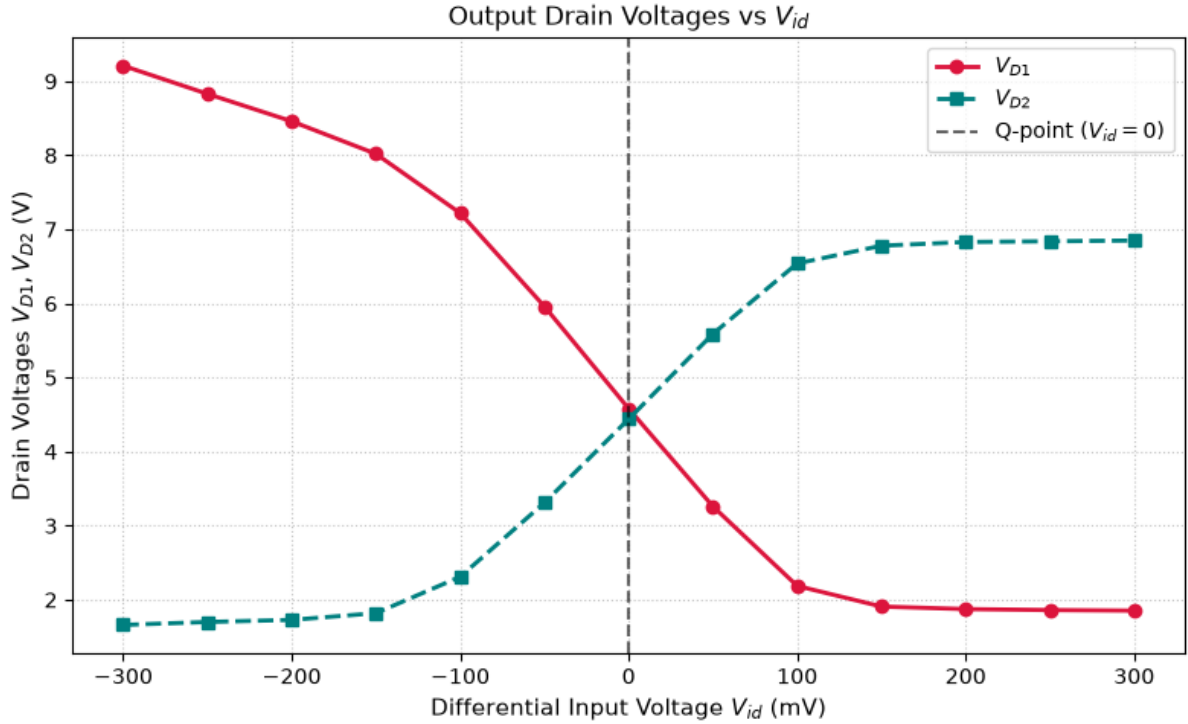


Figure 12: Experimental plot

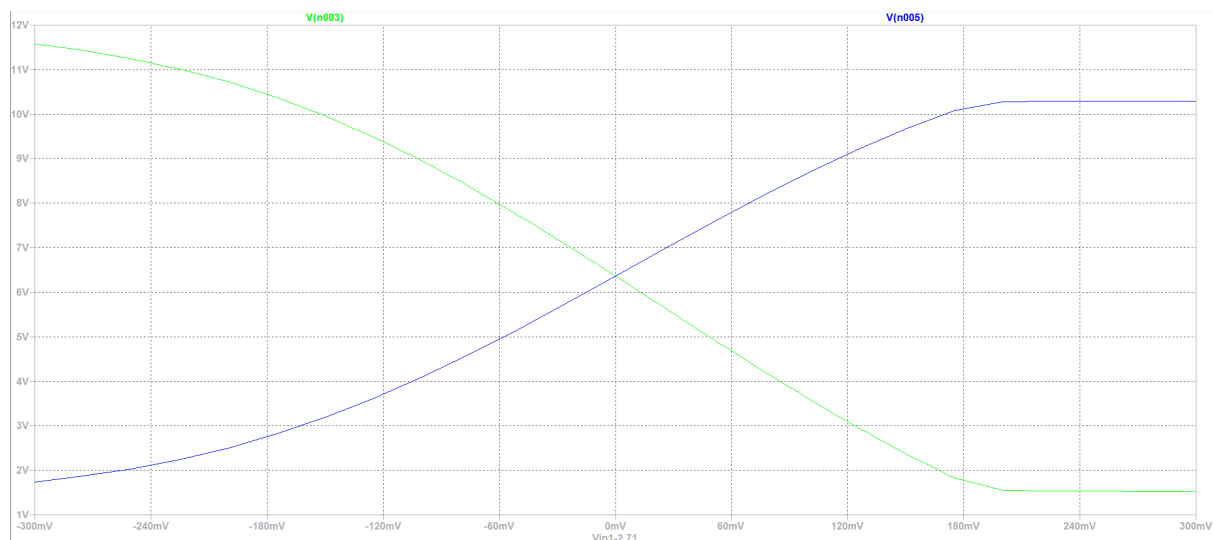


Figure 13: Simulation plot

6.4.2 Linear Scale Transfer Characteristic

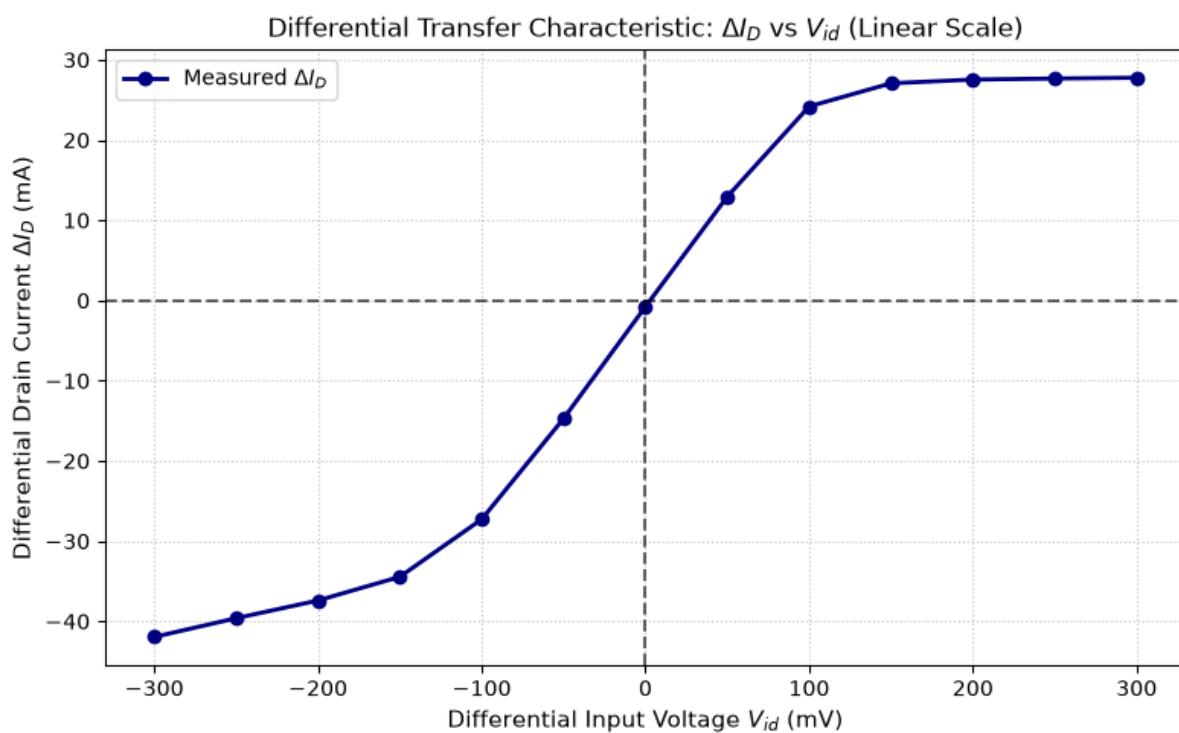


Figure 14: Experimental plot

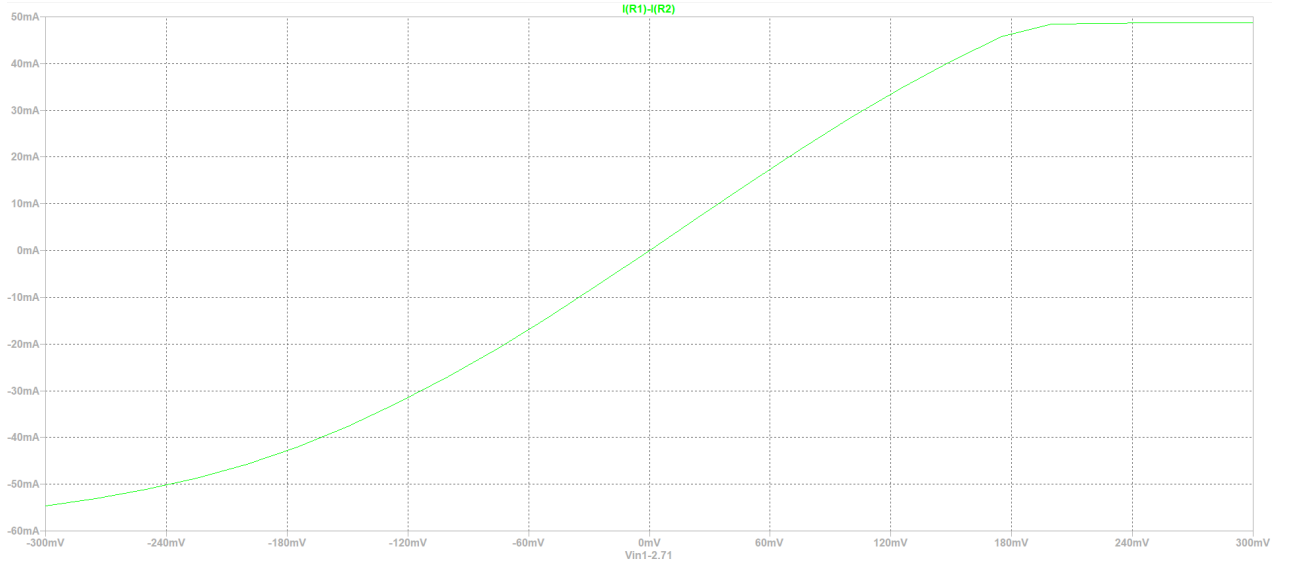


Figure 15: Simulation plot

6.4.3 Semi-Log Characteristic

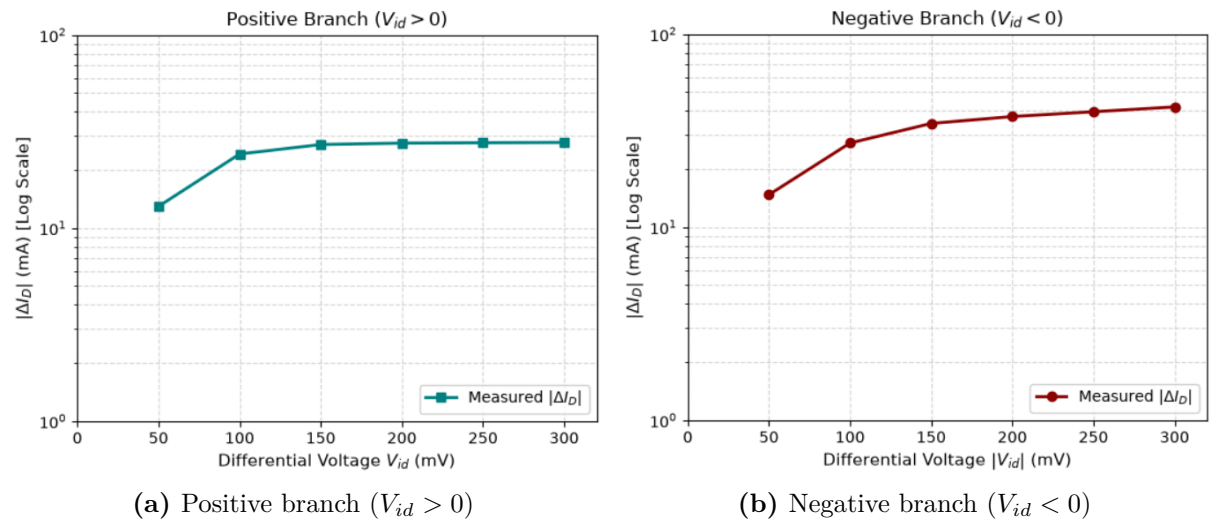


Figure 16: Experimental Plots

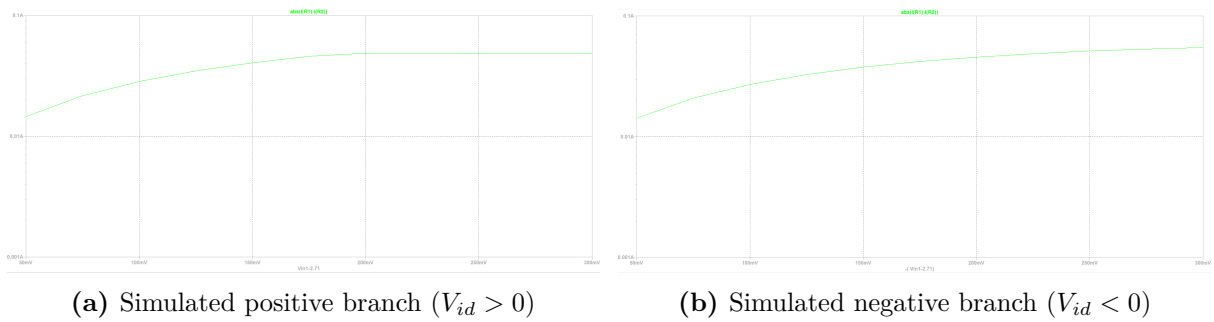


Figure 17: Simulation Plots

6.5 Calculations and Analysis

The differential-mode measurements were performed by keeping V_{in-} fixed at the quiescent common-mode voltage and varying V_{in+} . The differential input voltage is defined as

$$V_{id} = V_{in+} - V_{in-}. \quad (15)$$

For the experiment,

$$V_{in-} = V_{cm} = 2.71 \text{ V}. \quad (16)$$

The drain currents were calculated from the measured drain voltages using

$$I_{D1} = \frac{V_{DD} - V_{D1}}{R_D}, \quad I_{D2} = \frac{V_{DD} - V_{D2}}{R_D}, \quad (17)$$

where

$$V_{DD} = 12 \text{ V}, \quad R_D = 180 \text{ } \Omega. \quad (18)$$

The differential drain current is

$$\boxed{\Delta I_D = I_{D1} - I_{D2}} \quad (19)$$

6.5.1 Extraction of Slope Factor n

For weak inversion, the differential-mode characteristic is given by

$$\boxed{\Delta I_D = I_{SS} \tanh \left(\frac{V_{id}}{2nV_T} \right)} \quad (20)$$

where

$$V_T = \frac{kT}{q} \approx 25.85 \text{ mV} \quad (21)$$

at room temperature.

In the region where the current is still growing with $|V_{id}|$, the characteristic is approximately exponential,

$$|\Delta I_D| \propto \exp \left(\frac{|V_{id}|}{2nV_T} \right). \quad (22)$$

Hence, the slope m of the $\ln |\Delta I_D|$ versus $|V_{id}|$ line is

$$m = \frac{1}{2nV_T} \quad (23)$$

and therefore

$$\boxed{n = \frac{1}{2mV_T}} \quad (24)$$

The slope m is taken between $|V_{id}| = 50$ mV and $|V_{id}| = 150$ mV on each side, i.e. between the first and third measured points of the positive and negative branches:

$$m = \frac{\ln |\Delta I_D(150 \text{ mV})| - \ln |\Delta I_D(50 \text{ mV})|}{0.150 - 0.050}. \quad (25)$$

Using the measured semi-log slopes over this range,

$$m_- \approx 8.50 \text{ V}^{-1}, \quad m_+ \approx 7.38 \text{ V}^{-1}, \quad (26)$$

the corresponding slope factors are

$$n_- \approx 2.28, \quad n_+ \approx 2.62. \quad (27)$$

Thus,

$$\boxed{n_{\text{exp}} \approx 2.45 \quad (50\text{--}150 \text{ mV})} \quad (28)$$

Since $\ln |\Delta I_D|$ is not strictly linear in $|V_{id}|$ (the tanh saturates and the drains enter triode beyond ≈ 100 mV), this is an apparent slope factor that depends on the fitting range

6.5.2 Tail Current at the Q-Point

At the quiescent operating point,

$$V_{RSS} \approx 1.80 \text{ V}. \quad (29)$$

Since

$$R_{SS} = 22.4 \text{ } \Omega, \quad (30)$$

the tail current is

$$I_{SS} = \frac{V_{RSS}}{R_{SS}} = \frac{1.80}{22.4}. \quad (31)$$

Hence,

$$\boxed{I_{SS} \approx 80.36 \text{ mA}} \quad (32)$$

which is close to the intended operating current of approximately 80 mA.

6.5.3 Experimental vs Theoretical Tanh Curve

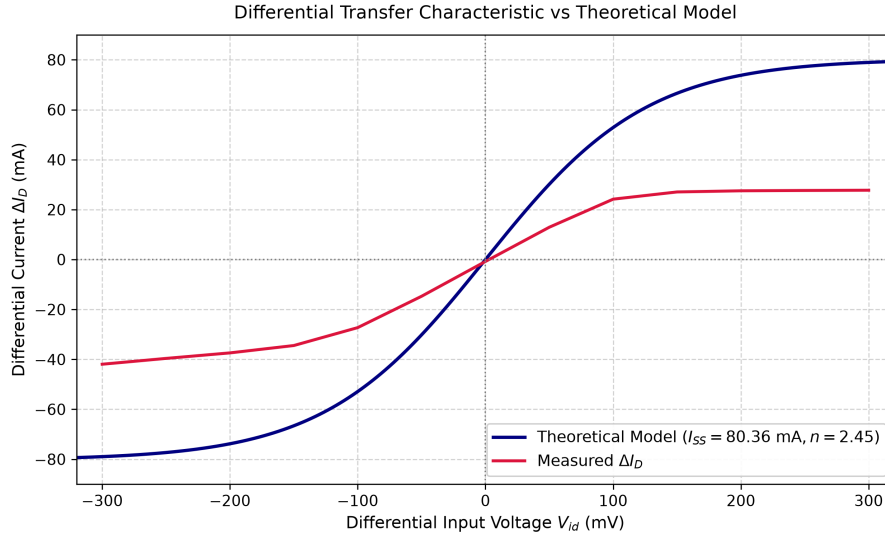


Figure 18: Comparison of the experimental and theoretical differential transfer characteristics

The experimental characteristic exhibits the expected nonlinear tanh-like behaviour, but with a smaller and asymmetric current swing compared to the theoretical model. This deviation can be attributed to practical non-idealities such as MOSFET parameter mismatch, non-ideal tail-current generation, and finite output resistance. The non-zero value of ΔI_D at $V_{id} = 0$ and the unequal positive and negative saturation levels indicate mismatch between the two transistor branches. Thus, while the theoretical model captures the overall differential-pair behaviour, significant quantitative deviation is observed in the experimental characteristic.

6.5.4 Experimental Transconductance

The small-signal differential transconductance is obtained from the slope of the ΔI_D versus V_{id} characteristic near the origin.

Using the points at $V_{id} = -50$ mV and $+50$ mV,

$$g_{m,\text{exp}} = \frac{12.94 - (-14.72)}{0.050 - (-0.050)} \times 10^{-3}. \quad (33)$$

Therefore,

$$\boxed{g_{m,\text{exp}} \approx 0.277 \text{ S}} \quad (34)$$

6.5.5 Theoretical Transconductance

Using the weak-inversion model,

$$g_{m,\text{theory}} = \frac{I_{SS}}{2nV_T}. \quad (35)$$

The factor of 2 arises because the differential pair consists of two matched transistors sharing the total tail current I_{SS} . At $V_{id} = 0$, the tail current is equally divided between the two transistors, giving $I_{D1} = I_{D2} = I_{SS}/2$. Hence, the differential current relation contains the factor 2 in the denominator:

$$\Delta I_D = I_{SS} \tanh\left(\frac{V_{id}}{2nV_T}\right)$$

Using

$$I_{SS} = 80.36 \text{ mA}, \quad n = 2.45, \quad V_T = 25.85 \text{ mV}, \quad (36)$$

gives

$$g_{m,\text{theory}} = \frac{0.08036}{2(2.45)(25.85 \times 10^{-3})}. \quad (37)$$

Therefore,

$$\boxed{g_{m,\text{theory}} \approx 0.634 \text{ S}} \quad (38)$$

6.5.6 Differential Gain

The differential voltage gain is approximately

$$A_d \approx g_m R_D. \quad (39)$$

Using the experimental transconductance,

$$A_{d,\text{exp}} = (0.277)(180). \quad (40)$$

Hence,

$$\boxed{A_{d,\text{exp}} \approx 49.8 \text{ V/V}} \quad (41)$$

Using the theoretical transconductance,

$$A_{d,\text{theory}} = (0.634)(180) \quad (42)$$

and therefore

$$\boxed{A_{d,\text{theory}} \approx 114.12 \text{ V/V}} \quad (43)$$

6.5.7 Comparison of Experimental and Theoretical Results

The experimental and theoretical values are summarized below.

Table 6: Comparison of experimental and theoretical results

Parameter	Experimental	Theoretical
Slope factor n	2.45	—
Transconductance g_m	0.277 S	0.634 S
Differential gain A_d	49.8 V/V	114.12 V/V

$$\begin{aligned} \text{Percentage Deviation in } g_m &= \frac{|g_{m,\text{theory}} - g_{m,\text{exp}}|}{g_{m,\text{theory}}} \times 100\% \\ &= \frac{|0.634 - 0.277|}{0.634} \times 100\% \approx 56.31\%. \end{aligned}$$

The experimental differential characteristic shows the expected operation of the MOSFET differential pair. For negative V_{id} , M_2 carries a larger fraction of the tail current, giving a negative ΔI_D . As V_{id} becomes positive, the current shifts towards M_1 , making ΔI_D positive.

At $V_{id} = 0$, the measured differential current is approximately

$$\boxed{\Delta I_D \approx -0.78 \text{ mA}} \quad (44)$$

rather than exactly zero, indicating a small mismatch between the two MOSFET branches.

For large positive and negative values of V_{id} , the differential current approaches a limiting value as most of the tail current is transferred to one transistor.

The difference between the experimental and theoretical transconductance is mainly due to the non-ideal behaviour of the practical MOSFET pair and deviations from the simplified weak-inversion model.

7 Common-Mode Rejection Ratio (CMRR)

7.1 Theoretical Background

For common-mode operation, the same voltage is applied to both input terminals of the differential amplifier:

$$V_{in+} = V_{in-} = V_{cm} \quad (45)$$

Hence,

$$V_{id} = V_{in+} - V_{in-} = 0. \quad (46)$$

The common-mode gain of the differential amplifier is approximately

$$A_{cm} \approx -\frac{R_D}{2R_{SS}} \quad (47)$$

For the present circuit,

$$R_D = 180 \, \Omega, \quad R_{SS} = 22.4 \, \Omega. \quad (48)$$

Therefore,

$$A_{cm, \text{theory}} = -\frac{180}{2(22.4)} \quad (49)$$

giving

$$A_{cm, \text{theory}} \approx -4.02 \, \text{V/V} \quad (50)$$

7.2 Experimental Common-Mode Characteristic

The common-mode characteristic was obtained by setting

$$V_{in+} = V_{in-} = V_{cm} \quad (51)$$

and varying V_{cm} .

The measured drain voltages and voltage across the tail resistor are shown in Table 7.

Table 7: Experimental common-mode measurements

V_{cm} (V)	V_{D1} (V)	V_{D2} (V)	V_{RSS} (V)	I_{SS} (mA)
0.0	11.94	12.00	0.001 mV	0.000045
0.9	11.94	12.00	0.450 mV	0.0201
1.0	11.10	10.94	0.232	10.36
1.5	9.35	8.98	0.683	30.49
2.0	7.50	7.10	1.12	50.00
2.5	5.55	5.26	1.58	70.54
2.71	4.59	4.43	1.80	80.36
3.0	3.463	3.460	2.05	91.52
3.5	2.332	2.343	2.323	103.71
4.0	2.326	2.336	2.332	104.11
4.5	2.318	2.330	2.323	103.71
5.0	2.317	2.329	2.323	103.71

The quiescent operating point selected for the differential-mode experiment was

$$\boxed{V_{cm} = 2.71 \text{ V}} \quad (52)$$

At this point,

$$V_{D1} = 4.59 \text{ V}, \quad V_{D2} = 4.43 \text{ V}. \quad (53)$$

The corresponding tail current is

$$I_{SS} = \frac{V_{RSS}}{R_{SS}} = \frac{1.80}{22.4} \quad (54)$$

and hence

$$\boxed{I_{SS} \approx 80.36 \text{ mA}} \quad (55)$$

The small difference between the two drain voltages at the quiescent point indicates a mismatch between the two MOSFET branches.

7.3 Observed Common-Mode Range

The drain voltages vary considerably at low values of V_{cm} . With increasing V_{cm} , the drain voltages become approximately constant.

From the measured data, an approximately flat region is observed between

$$\boxed{3.5 \text{ V} \lesssim V_{cm} \lesssim 5.0 \text{ V}} \quad (56)$$

In this region,

$$V_{D1} \approx 2.32 \text{ V}, \quad V_{D2} \approx 2.33 \text{ V}. \quad (57)$$

This is the experimentally observed flat region of the common-mode characteristic; it does not necessarily represent the complete theoretical ICMR of the amplifier.

7.4 Average Drain Voltage

The average drain voltage is defined as

$$\boxed{V_D^{(\text{avg})} = \frac{V_{D1} + V_{D2}}{2}} \quad (58)$$

For the approximately flat region, the measured values are

Table 8: Average drain voltage in the approximately flat region

V_{cm} (V)	V_{D1} (V)	V_{D2} (V)	$V_D^{(\text{avg})}$ (V)
3.5	2.332	2.343	2.3375
4.0	2.326	2.336	2.3310
4.5	2.318	2.330	2.3240
5.0	2.317	2.329	2.3230

7.5 Experimental Common-Mode Gain

The experimental common-mode response is obtained from the slope of the average drain voltage with respect to V_{cm} :

$$A_{cm,\text{exp}} = \frac{\Delta V_D^{(\text{avg})}}{\Delta V_{cm}}. \quad (59)$$

Using the endpoints of the approximately flat region,

$$A_{cm,\text{exp}} = \frac{2.3230 - 2.3375}{5.0 - 3.5} \quad (60)$$

Therefore,

$$\boxed{A_{cm,\text{exp}} \approx -9.67 \times 10^{-3} \text{ V/V}} \quad (61)$$

7.6 Calculation of CMRR

The common-mode rejection ratio is defined as

$$\boxed{\text{CMRR} = \left| \frac{A_d}{A_{cm}} \right|} \quad (62)$$

From the differential-mode experiment,

$$A_{d,\text{exp}} \approx 49.8 \text{ V/V}. \quad (63)$$

Using

$$A_{cm,\text{exp}} \approx -9.67 \times 10^{-3} \text{ V/V}, \quad (64)$$

the experimental CMRR is

$$\text{CMRR} = \left| \frac{49.8}{-9.67 \times 10^{-3}} \right| \quad (65)$$

giving

$$\boxed{\text{CMRR} \approx 5.15 \times 10^3} \quad (66)$$

In decibels,

$$\text{CMRR}_{dB} = 20 \log_{10}(\text{CMRR}) \quad (67)$$

and hence

$$\boxed{\text{CMRR}_{dB} \approx 74.2 \text{ dB}} \quad (68)$$

7.7 Comparison of Results

The theoretical and experimental common-mode gains are

$$\boxed{A_{cm,\text{theory}} \approx -4.02 \text{ V/V}} \quad (69)$$

and

$$\boxed{A_{cm,\text{exp}} \approx -9.67 \times 10^{-3} \text{ V/V}.} \quad (70)$$

The difference between the two values is due to the non-ideal behaviour of the MOS-FETs and the limitations of the simplified common-mode model.

The measured characteristic shows an approximately flat common-mode response over the range 3.5–5.0 V, while the experimentally obtained CMRR is approximately

$$\boxed{\text{CMRR}_{dB} \approx 74.2 \text{ dB.}} \quad (71)$$

8 Comparison of Results and Error Analysis

8.1 Summary of Experimental Results

The principal experimentally obtained quantities are summarized in Table 9.

Table 9: Summary of experimental results

Quantity	Experimental Value
g_m	0.277 S
A_d	49.8 V/V
A_{cm}	-9.67×10^{-3} V/V
CMRR	5.15×10^3
CMRR	74.2 dB
Observed flat common-mode region	3.5–5.0 V
Q-point V_{cm}	2.71 V
Q-point V_{D1}	4.58 V
Q-point V_{D2}	4.44 V
Q-point mismatch	0.14 V

8.2 Comparison of Theoretical and Experimental Values

For the present circuit,

$$R_D = 180 \, \Omega, \quad R_{SS} = 22.4 \, \Omega. \quad (72)$$

The theoretical common-mode gain is

$$A_{cm,\text{theory}} = -\frac{R_D}{2R_{SS}} = -\frac{180}{2(22.4)} \quad (73)$$

giving

$$\boxed{A_{cm,\text{theory}} \approx -4.02 \text{ V/V}} \quad (74)$$

The experimentally obtained common-mode gain is

$$\boxed{A_{cm,\text{exp}} \approx -9.67 \times 10^{-3} \text{ V/V}} \quad (75)$$

The difference between the two values arises because the theoretical expression is based on a simplified differential-pair model, whereas the practical circuit includes device mismatch, resistor tolerances, and other non-ideal effects.

8.3 Differential-Mode Results

The experimental transconductance obtained from the slope of the ΔI_D versus V_{id} characteristic near the origin is

$$\boxed{g_{m,\text{exp}} \approx 0.277 \text{ S}} \quad (76)$$

The corresponding differential gain is

$$A_{d,\text{exp}} = g_{m,\text{exp}} R_D \quad (77)$$

and therefore

$$\boxed{A_{d,\text{exp}} \approx 49.8 \text{ V/V}} \quad (78)$$

The experimental differential characteristic shows the expected operation of a differential pair.

For negative V_{id} ,

$$I_{D2} > I_{D1} \quad (79)$$

and hence

$$\Delta I_D < 0. \quad (80)$$

For positive V_{id} ,

$$I_{D1} > I_{D2} \quad (81)$$

and hence

$$\Delta I_D > 0. \quad (82)$$

Thus, increasing V_{id} transfers current from the M_2 branch to the M_1 branch.

8.4 Differential Transfer Characteristic

The weak-inversion differential transfer characteristic is

$$\boxed{\Delta I_D = I_{SS} \tanh \left(\frac{V_{id}}{2nV_T} \right)} \quad (83)$$

The experimental characteristic follows the expected general trend: ΔI_D changes from negative to positive as V_{id} is increased and approaches a limiting value for large $|V_{id}|$.

At the nominally zero differential input,

$$V_{id} = 0, \quad (84)$$

the measured differential current is approximately

$$\boxed{\Delta I_D \approx -0.78 \text{ mA}} \quad (85)$$

instead of exactly zero. This indicates a small mismatch between the two MOSFET branches.

8.5 Transistor Mismatch

At the quiescent operating point,

$$V_{D1} = 4.59 \text{ V}, \quad V_{D2} = 4.43 \text{ V}. \quad (86)$$

Hence,

$$\boxed{V_{D1} - V_{D2} = 0.16 \text{ V}} \quad (87)$$

Using $R_D = 180 \text{ } \Omega$, the corresponding drain currents are

$$I_{D1} = \frac{12 - 4.59}{180} \approx 41.17 \text{ mA} \quad (88)$$

and

$$I_{D2} = \frac{12 - 4.43}{180} \approx 42.06 \text{ mA}. \quad (89)$$

Therefore,

$$\boxed{\Delta I_D = I_{D1} - I_{D2} \approx -0.89 \text{ mA}} \quad (90)$$

The non-zero differential current at $V_{id} = 0$ is consistent with mismatch between the two MOSFET branches.

8.6 Common-Mode Rejection

From the common-mode measurements, the average drain voltage changes only slightly between $V_{cm} = 3.5 \text{ V}$ and 5.0 V .

At $V_{cm} = 3.5 \text{ V}$,

$$V_D^{(\text{avg})} = 2.3375 \text{ V} \quad (91)$$

whereas at $V_{cm} = 5.0 \text{ V}$,

$$V_D^{(\text{avg})} = 2.3230 \text{ V.} \quad (92)$$

Thus,

$$\Delta V_D^{(\text{avg})} = 2.3230 - 2.3375 = -0.0145 \text{ V.} \quad (93)$$

For a common-mode input change of

$$\Delta V_{cm} = 5.0 - 3.5 = 1.5 \text{ V,} \quad (94)$$

the experimental common-mode gain is

$$\boxed{A_{cm,\text{exp}} \approx -9.67 \times 10^{-3} \text{ V/V}} \quad (95)$$

The small change in drain voltage over this region demonstrates the common-mode rejection capability of the differential amplifier.

8.7 Sources of Experimental Error

The main sources of discrepancy between the theoretical and experimental results are:

- Mismatch between the two MOSFETs in the differential pair.
- Tolerance of the drain and tail resistors R_D and R_{SS} .
- Measurement uncertainty in V_{D1} and V_{D2} .
- Finite output resistance and other non-ideal MOSFET characteristics.
- Deviation of the practical MOSFET characteristics from the simplified theoretical model.

8.8 Discussion

The experimental results demonstrate the expected operation of the MOSFET differential amplifier. In differential mode, varying V_{id} transfers current between the two branches, causing ΔI_D to change from negative to positive.

In common-mode operation, the drain voltages remain nearly constant over the observed flat region, indicating common-mode rejection. The non-zero ΔI_D at $V_{id} = 0$ and unequal drain voltages indicate a small mismatch between the MOSFET branches.

9 Precautions

- Verify all MOSFET pin connections using the datasheet before powering the circuit.
- Keep the V_{DS} and V_{GS} values within the specified device limits.
- Keep the power supply current limit enabled throughout the experiment.
- Switch off the power supply before making any circuit changes.
- Vary the input voltages gradually to avoid excessive current or voltage.
- Record measurements only after the circuit reaches a stable operating condition.

10 Conclusion

A two-transistor MOSFET differential amplifier using the SI9926BDY dual N-channel MOSFET was experimentally studied in differential and common-mode operation. The quiescent point was obtained at $V_{cm} = 2.71$ V, with $V_{D1} = 4.58$ V and $V_{D2} = 4.44$ V, indicating a small mismatch between the two MOSFET branches.

From the differential-mode characteristic,

$$g_{m,\text{measured}} \approx 0.277 \text{ S}, \quad A_{d,\text{measured}} \approx 49.8 \text{ V/V},$$

and the extracted slope factor was

$$n_{\text{measured}} \approx 2.45.$$

Using this value, the theoretical transconductance was approximately 0.634 S, giving a deviation of about 56.31%.

For common-mode operation, the drain voltages remained nearly constant over the measured 3.5–5.0 V region, giving

$$A_{cm,\text{measured}} \approx -9.67 \times 10^{-3} \text{ V/V}, \quad \text{CMRR} \approx 5.15 \times 10^3 \approx 74.2 \text{ dB}.$$

Overall, the experiment demonstrated the expected differential and common-mode behaviour, with deviations arising from MOSFET mismatch, resistor tolerances, measurement uncertainty, and practical device non-idealities.